UNIX Past

A look at the key moments and history behind the UNIX operating system.

> ...a punning colleague called UNICS... an 'emasculated Multics'.
>
> A Brief Look at the Early History

> Unix was born in 1969... this paper presents a technical and social history of the evolution of the system.
>
> Dennis M. Ritchie

> ...the number of UNIX installations has grown to 10, with more expected...
>
> Ritchie & Thompson, 1972

Since escaping AT&T's Bell Laboratories in the early 1970s, the UNIX system's influence has been immense. Universities, research institutes, and companies used it to pioneer technologies that are now essential—from CAD and lab simulations to the very foundation of the Internet.

A Fragmented Landscape

By the late 1970s, graduates who had used UNIX systems in university labs entered the workforce and demanded it. In

response, vendors marketed their own diverging, incompatible versions of the system. The UNIX® trademark was ubiquitous, but it was applied to a multitude of different products that couldn't work together.

The Rise of Open Systems

Concerned by market fragmentation, a group of vendors formed X/Open in 1984 to define a comprehensive "open systems" environment based on agreed-upon standards. They chose the UNIX system as the foundation, creating a clear industry need for a single, standard version. The question was: which one?

> *Open systems were those that would meet agreed specifications or standards. This resulted in the formation of X/Open Company Ltd whose remit was, and today in the guise of The Open Group remains, to define a comprehensive open systems environment.*

The UNIX Wars

In 1987, AT&T and Sun Microsystems formed a pact to unify the market around their version. Fearing this alliance, the rest of the industry formed the competing Open Software Foundation (OSF). The resulting "UNIX Wars" divided the community between AT&T's UNIX System V and the OSF's OSF/1, while X/Open held the neutral center ground.

Towards a Unified Standard

In a pivotal move, Novell (which had acquired AT&T's UNIX System Laboratories) transferred the rights to the UNIX trademark to the vendor-neutral X/Open in 1993. This separated the brand from any single company's code. By 1995, X/Open introduced the UNIX 95 brand, guaranteeing that certified systems met the Single UNIX Specification—a mission The Open Group continues today.

Timeline of a Revolution

Key milestones in the history of the UNIX system.

1969 The Beginning

UNIX history starts back in 1969, when Ken Thompson, Dennis Ritchie and others started working on the "little-used PDP-7 in a corner" at Bell Labs.

1971 First Edition

It had an assembler for a PDP-11/20, file system, fork(), roff and ed. It was used for text processing of patent documents.

1973 Fourth Edition

It was rewritten in C. This made it portable and changed the history of OS's.

1975 Sixth Edition

Also known as Version 6, this is the first to be widely available out side of Bell Labs. The first BSD version (1.x) was derived from V6.

1979 Seventh Edition

It was a "improvement over all preceding and following Unices" [Bourne]. It had C,

UUCP and the Bourne shell. It was ported to the VAX.

1980 Xenix

Microsoft introduces Xenix. 32V and 4BSD introduced.

1982 System III

AT&T's UNIX System Group (USG) release System III, the first public release outside Bell Laboratories. SunOS 1.0 ships. HP-UX introduced.

1983 System V

AT&T announces UNIX System V, the first supported release. Installed base 45,000.

1984 4.2BSD

University of California at Berkeley releases 4.2BSD, includes TCP/IP, new signals and much more. X/Open formed.

1984 SVR2

System V Release 2 introduced. At this time there are 100,000 UNIX installations around the world.

1986 4.3BSD

4.3BSD released, including internet name server. SVID introduced. NFS shipped. AIX announced. Installed base 250,000.

1987 SVR3

System V Release 3 including STREAMS, TLI, RFS. At this time there are 750,000 UNIX installations around the world.

1988 POSIX.1

POSIX.1 published. Open Software Foundation (OSF) and UNIX International (UI) formed.

1989 System V

AT&T UNIX Software Operation formed in preparation for spinoff of USL.

1989 SVR4

UNIX System V Release 4 ships, unifying System V, BSD and Xenix. Installed base 1.2 million.

1990 XPG3

X/Open launches XPG3 Brand. OSF/1 debuts. Plan 9 from Bell Labs ships.

1991 Linux

UNIX System Laboratories (USL) becomes a company. Linus Torvalds commences Linux development. Solaris 1.0 debuts.

1992 SVR4.2

USL releases UNIX System V Release 4.2 (Destiny). XPG4 Brand launched. Novell announces intent to acquire USL.

1993 4.4BSD

4.4BSD the final release from Berkeley. Novell acquires USL.

Late 1993 SVR4.2MP

Novell transfers rights to the "UNIX" trademark and the Single UNIX Specification to X/Open. COSE initiative delivers "Spec 1170".

1994 Single UNIX Spec

X/Open introduces the Single UNIX Specification (formerly Spec 1170), separating the trademark from any code stream.

1995 UNIX 95

X/Open introduces the UNIX 95 branding programme. Novell sells UnixWare business line to SCO.

1996 The Open Group

The Open Group forms as a merger of OSF and X/Open.

1997 Version 2

Version 2 of the Single UNIX Specification is released, adding support for realtime, threads and 64-bit processors.

1998 UNIX 98

The Open Group introduces the UNIX 98 family of brands (Base, Workstation, Server).

1999 UNIX at 30

The UNIX system reaches its 30th anniversary. Joint development of a revision to POSIX and the Single UNIX Spec begins.

2001 Version 3

Version 3 of the Single UNIX Specification unites IEEE POSIX, The Open Group and the industry efforts.

2003 ISO/IEC 9945:2003

The core volumes of Version 3 are approved as an international standard.

2007 Mac OS X Certified

Apple Mac OS X is certified to UNIX 03.

2008 ISO/IEC 9945:2008

The latest revision of the UNIX API set is formally standardized at ISO/IEC, IEEE and The Open Group.

2009 UNIX at 40

IDC reports the UNIX market at $69 billion in 2008, predicting $74 billion in 2013.

2010 Desktop Dominance

Apple reports 50 million desktops and growing—all Certified UNIX systems.

2015 Version 4

The Single UNIX Specification Version 4 (SUSv4) is published, forming the basis for the UNIX V7 Certification program.

2018 First UNIX V7 Cert

Oracle Solaris 11.4 becomes the first product to be certified as conforming to the latest UNIX V7 standard.

2019 UNIX at 50

Happy 50th Birthday, UNIX!

2020 IBM certifies AIX to the UNIX V7 standard

Increased reliance on POSIX and UNIX systems during the shift to remote working during the pandemic.

2022 Apple certifies macOS 12 Monterey

Apple continues its long-standing commitment by certifying its annual release for the 15th year. Estimated sales of Mac computers exceed 200 million units.

2024 25 Years of the Austin Group

25 years of joint development of the POSIX and Single UNIX Spec. The Single UNIX Specification Version 5 (SUSv5) is published.

2025 Apple certifies the macOS Tahoe release

macOS continues as the largest installed base of any certified UNIX platform - estimated at tens of millions of active systems.

2026 ISO/IEC 9945-2026 Published

The core volumes of the Single UNIX Specification Version 5 are published by ISO/IEC.

A Spec with No Name

How a cowboy legend helped unify the UNIX world.

The Problem: Fragmentation

In the early 1990s, major vendors like Sun, HP, and IBM had developed incompatible versions of UNIX, making it hard for software to be portable across platforms.

A Spec with No Name

A unified standard was needed, but the effort was politically sensitive. As a result, it was informally called "The Spec with No Name," later known as Spec 1170.

Enter: Eastwood

To ensure vendors implemented the spec correctly, a conformance test suite was nicknamed "Eastwood"— an

inside joke referencing Clint Eastwood's character, "The Man with No Name."

A Lasting Legacy

This effort laid the foundation for The Single UNIX Specification and was a key part of the UNIX brand consolidation.

> *"This is the Unix philosophy: Write programs that do one thing and do it well. Write programs to work together. Write programs to handle text streams, because that is a universal interface."*
>
> Attributed to *Doug McIlroy*

An Enduring Legacy

While the corporate battles have faded, the core philosophy and UNIX technical innovations are more relevant than ever. They form the bedrock of nearly all modern computing, from the servers that run the internet (Linux), to the phones in our pockets (Android/iOS), to the computers on our desks (macOS, Windows Subsystem for Linux).